LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Ch 4 Test: Carbon and its Compounds

Chapter / Topic: Carbon and its Compounds

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Write balanced chemical equations and structural formulae wherever asked.

SECTION A

6 marks

Choose the correct option.

Q1. Ethane, with the molecular formula C_2H_6, has: **[1]**

(a) 6 covalent bonds (b) 7 covalent bonds
(c) 8 covalent bonds (d) 9 covalent bonds

Q2. Butanone is a four-carbon compound with the functional group: **[1]**

(a) carboxylic acid (b) aldehyde
(c) ketone (d) alcohol

Q3. The next higher homologue of ethanol (C_2H_5OH) is: **[1]**

(a) CH_3OH (b) C_2H_5COOH
(c) C_3H_8 (d) C_3H_7OH

Q4. While cooking, if the bottom of the vessel is getting blackened on the outside, it means that: **[1]**

(a) the food is not cooked completely
(b) the fuel is not burning completely
(c) the fuel is wet
(d) the fuel is burning completely

Q5. Which of the following is an unsaturated hydrocarbon? **[1]**

(a) C_2H_4 (b) C_3H_8
(c) CH_4 (d) C_2H_6

Q6. **Assertion (A):** Carbon forms covalent bonds rather than ionic bonds.
Reason (R): Carbon has four valence electrons, and it is difficult for it to gain or lose four electrons. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. What is a homologous series? Write the formulae of the first two members of the alkyne series. **[2]**

OR

Give two reasons why carbon forms a very large number of compounds.

Q8. What is saponification? Write the balanced equation for the reaction of ethyl ethanoate with sodium hydroxide. **[2]**

SECTION C

6 marks

Q9. Answer the following about ethanol: **[3]**

(i) Write the balanced equation for the reaction of ethanol with sodium. **[1]**

(ii) What is formed when ethanol is heated with excess concentrated sulphuric acid at 443 K? Write the equation. **[1]**

(iii) Name the product formed when ethanol is oxidised with alkaline potassium permanganate. **[1]**

Q10. What is esterification? Write the balanced equation for the formation of an ester from ethanoic acid and ethanol, and give one use of esters. **[3]**

OR

How are soaps different from detergents? Why does soap not work well in hard water?

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) Explain the cleansing action of soap. Use the idea of micelles. **[3]**

(ii) What is hydrogenation? Give one industrial use. **[2]**

OR

(a) What are structural isomers? Write the structures of the two isomers of butane. [3]
(b) Why are covalent compounds generally poor conductors of electricity? [2]

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. Carbon compounds burn in oxygen to give carbon dioxide and water, along with heat and light. Saturated hydrocarbons generally burn with a clean blue flame, while unsaturated hydrocarbons give a yellow flame with lots of black smoke. In a gas stove, the air holes let in enough oxygen for complete combustion. If the air holes get blocked, the flame becomes yellow and sooty. **[4]**

(i) Write the balanced equation for the complete combustion of methane. **[1]**

(ii) Which type of hydrocarbon burns with a sooty yellow flame? **[1]**

(iii) Why does the flame become sooty when the air holes are blocked, and why is this a problem? **[2]**

~ All the best! ~